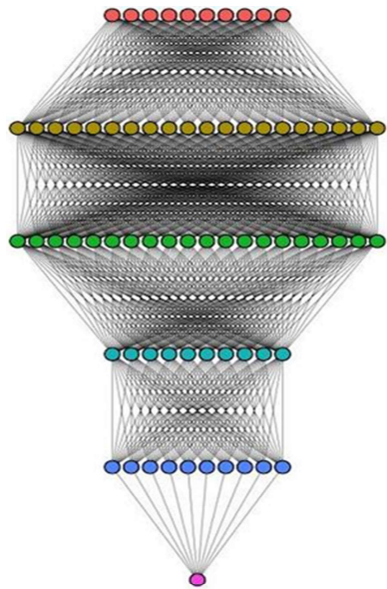


Multi-Layered Perceptron



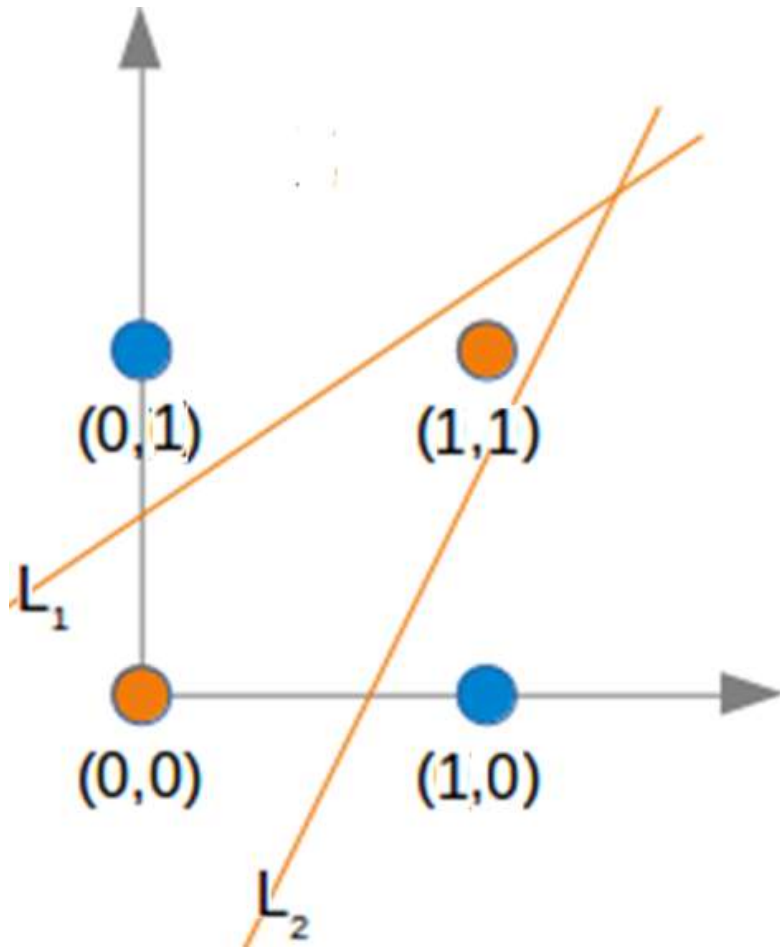
Prithwijit Guha
Dept. of EEE, IIT Guwahati

Supervised Learning



- Bayesian Classification, MAP, Chebyshev Inequality
- Performance Measures, Confusion Matrix, ROC Curves
- Logistic Regression
- Perceptron
- Multi-Layer Perceptron (MLP), ELM Learning
- MLP Architectures, Learning, Interpretations
- Non-parametric Methods and K-NN
- Radial Basis Function Neural Networks
- Data Balancing; SMOTE & Weighted Loss Functions
- Classification & Regression Trees
- Support Vector Machines & Multiple Kernel Learning
- Ensemble Methods, Bagging and Boosting

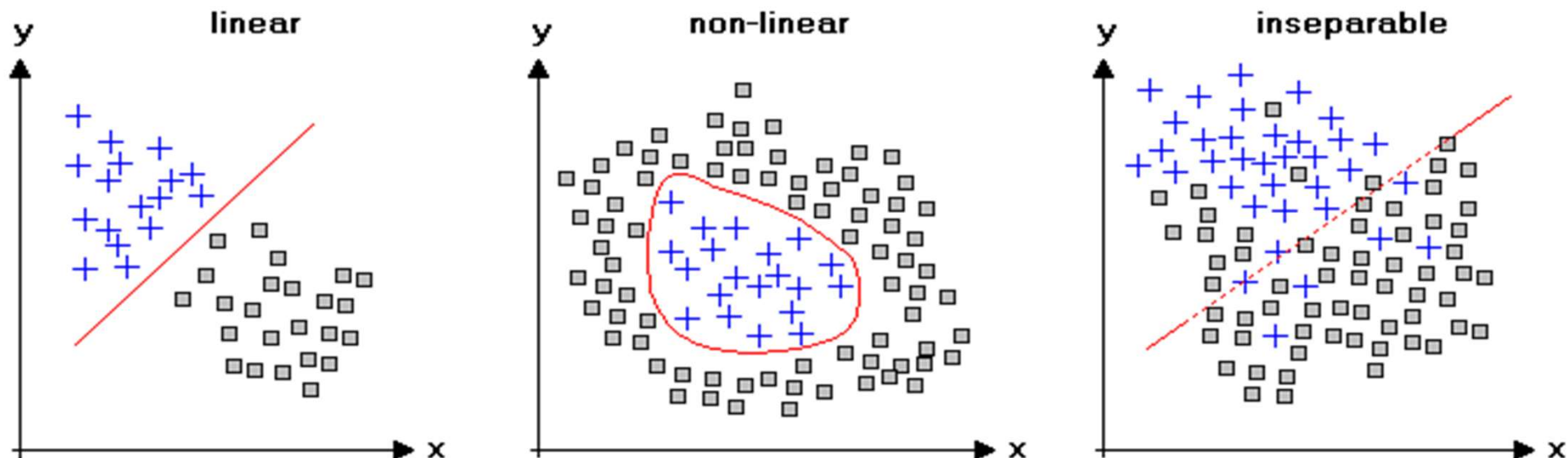
Perceptron: Failure Cases



The XOR Problem

$$y = x_1(1 - x_2) + x_2(1 - x_1)$$

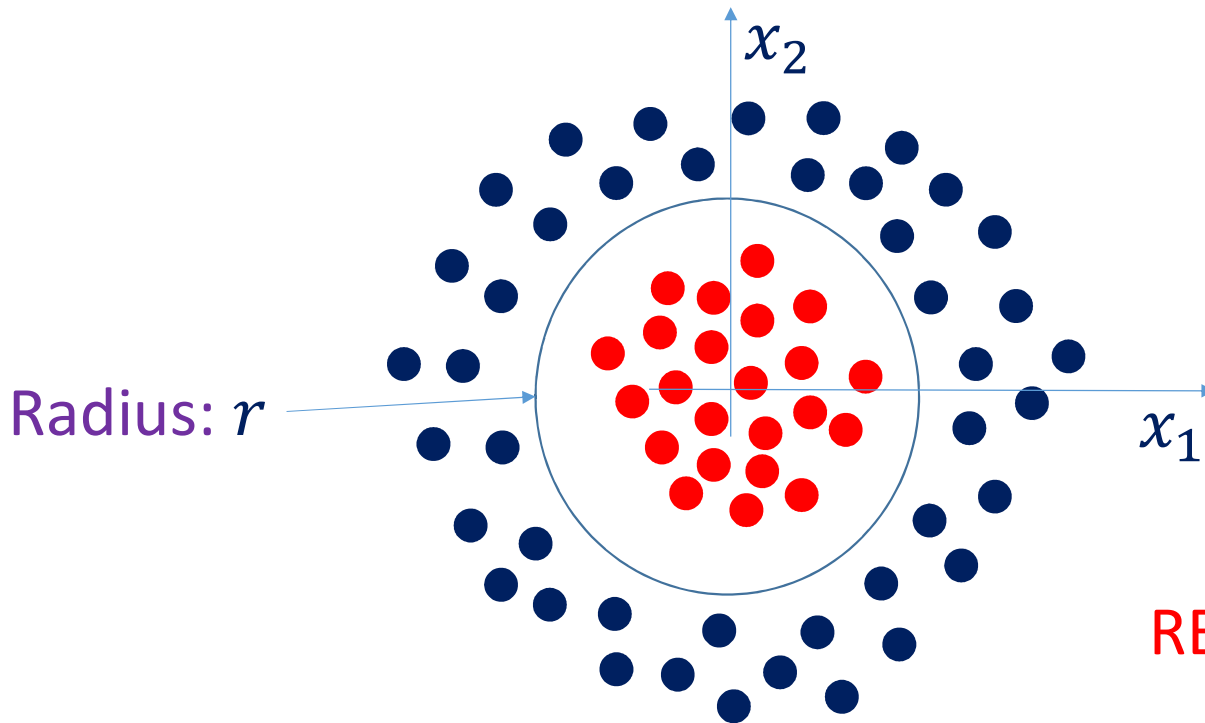
Perceptron: Failure Cases



Cover's Theorem of Separability

Nonlinear Transformation to a Higher Dimensional Space Increases the Probability of Linear Separability

The Feature Space: Not Linearly Separable



Feature: $\mathbf{x} \equiv (x_1, x_2)$

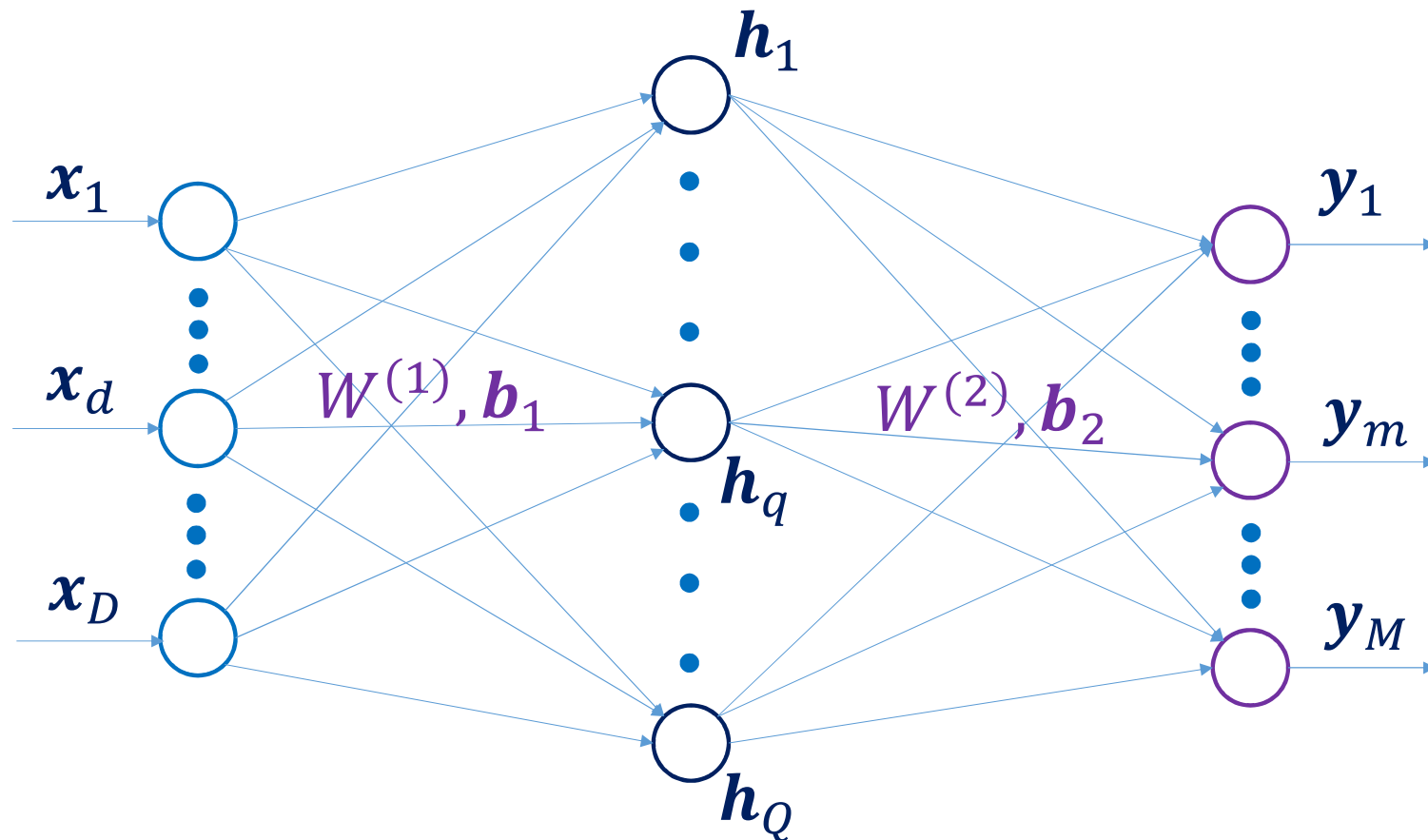
Define: $z \equiv r^2 - x_1^2 - x_2^2$

RED: $z > 0$

BLUE: $z < 0$

Transformed Feature: $\mathbf{y} = f(\mathbf{x}) = (x_1, x_2, z) \rightarrow$ Separable

Multi-Layered Perceptron (MLP) Architecture



Single Layer Feed Forward Network (SLFN)

Feed Forward through SLFN

$$\mathcal{S} = \{(\mathbf{x}_n, \mathbf{y}_n); n = 1, \dots, N\}$$

$$\mathbf{x} \in \mathbb{R}^D$$

$$\mathbf{y} \in \mathbb{R}^M$$

$$\mathbf{U}_{qn} = \mathbf{u}_n[q] = \sum_{d=1}^D W_{qd}^{(1)} \mathbf{x}_n[d] + \mathbf{b}_1[q]$$



$$\mathbf{u}_n = W^{(1)} \mathbf{x}_n + \mathbf{b}_1 \quad \longrightarrow \quad \mathbf{h}_n[\mathbf{q}] = g_1(\mathbf{u}_n[q])$$

From Input To Hidden Layer Activation

Feed Forward through SLFN

$$\mathbf{V}_{mn} = \mathbf{v}_n[m] = \sum_{q=1}^Q W_{mq}^{(2)} \mathbf{h}_n[q] + \mathbf{b}_2[m]$$



$$\mathbf{v}_n = W^{(2)} \mathbf{h}_n + \mathbf{b}_2$$



$$\mathbf{y}_n[\mathbf{m}] = g_2(\mathbf{v}_n[m])$$

From Hidden To Output Layer Activation

MLP: Problem

Consider a SLFN with $D = 2$ Inputs, $Q = 3$ Hidden Layer Nodes and $M = 2$ Outputs. Sigmoid Activation is used in both Hidden and Output Layers.

$$W^{(1)} = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix} \quad b_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}$$

$$X = \begin{bmatrix} -0.25 \\ 0.57 \end{bmatrix}$$

$$W^{(2)} = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix}$$

$$Y = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

MLP: Problem

Consider a SLFN with $D = 2$ Inputs, $Q = 3$ Hidden Layer Nodes and $M = 2$ Outputs. Sigmoid Activation is used in both Hidden and Output Layers.

$$W^{(1)} = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix}$$

$$\mathbf{b}_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}$$

$$X = \begin{bmatrix} -0.25 \\ 0.57 \end{bmatrix}$$

$$U = W^{(1)}X + \mathbf{b}_1$$

$$W^{(2)} = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix}$$

$$\mathbf{b}_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix}$$

$$U = \begin{bmatrix} 0.6615 \\ 0.0994 \\ -0.6914 \end{bmatrix}$$

MLP: Problem

Consider a SLFN with $D = 2$ Inputs, $Q = 3$ Hidden Layer Nodes and $M = 2$ Outputs. Sigmoid Activation is used in both Hidden and Output Layers.

$$W^{(1)} = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}$$

$$U = \begin{bmatrix} 0.6615 \\ 0.0994 \\ -0.6914 \end{bmatrix}$$

$$H[q] = \text{sigmoid}(U[q])$$

$$W^{(2)} = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix}$$

$$H = \begin{bmatrix} 0.6596 \\ 0.5248 \\ 0.3337 \end{bmatrix}$$

MLP: Problem

Consider a SLFN with $D = 2$ Inputs, $Q = 3$ Hidden Layer Nodes and $M = 2$ Outputs. Sigmoid Activation is used in both Hidden and Output Layers.

$$W^{(1)} = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}$$

$$H = \begin{bmatrix} 0.6596 \\ 0.5248 \\ 0.3337 \end{bmatrix}$$

$$V = W^{(2)}H + b_2$$

$$W^{(2)} = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.1390 \\ 0.9261 \end{bmatrix}$$

MLP: Problem

Consider a SLFN with $D = 2$ Inputs, $Q = 3$ Hidden Layer Nodes and $M = 2$ Outputs. Sigmoid Activation is used in both Hidden and Output Layers.

$$W^{(1)} = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.1390 \\ 0.9261 \end{bmatrix}$$

$$Y[m] = \text{sigmoid}(V[m])$$

$$W^{(2)} = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0.5347 \\ 0.7163 \end{bmatrix}$$

MLP: Parameters and Computations

Consider a SLFN with $D = 1000$ Input Nodes, $Q = 500$ Hidden Layer Nodes and $M = 20$ Outputs. Sigmoid Activation is used in Hidden Layer. The Output Layers use SoftMax Activation. Compute the Total Number of Learnable Parameters, Multiplications, Additions, Divisions and Exponential Evaluations

MLP: Total Learnable Parameters

$W^{(1)}: Q \times D$ Parameters

$W^{(2)}: M \times Q$ Parameters

$b_1: Q \times 1$ Parameters

$b_2: M \times 1$ Parameters

Total: $Q \times (M + D + 1) + M$ Parameters

$$D = 1000, Q = 500, M = 20$$

Total: $500 \times (20 + 1000 + 1) + 20 = 510520 \sim 0.51M$ Parameters

MLP: Computations

Fill up this Table

OPS	MUL	ADD	DIV	EXP
$\tilde{U} = W^{(1)}X$				
$U = \tilde{U} + \mathbf{b}_1$				
$H = \text{Sigmoid}(U)$				
$\tilde{V} = W^{(2)}H$				
$V = \tilde{V} + \mathbf{b}_2$				
$Y = \text{SoftMax}(V)$				
TOTAL				

MLP: Computations

OPS	MUL	ADD	DIV	EXP
$\tilde{U} = W^{(1)}X$	$Q \times D$	$Q \times D$	0	0
$U = \tilde{U} + \mathbf{b}_1$	0	Q	0	0
$H = \text{Sigmoid}(U)$	0	Q	Q	Q
$\tilde{V} = W^{(2)}H$	$M \times Q$	$M \times Q$	0	0
$V = \tilde{V} + \mathbf{b}_2$	0	M	0	0
$Y = \text{SoftMax}(V)$	0	M	M	M
TOTAL	$Q \times (D + M)$	$Q \times (D + M) + 2M$	$Q + M$	$Q + M$

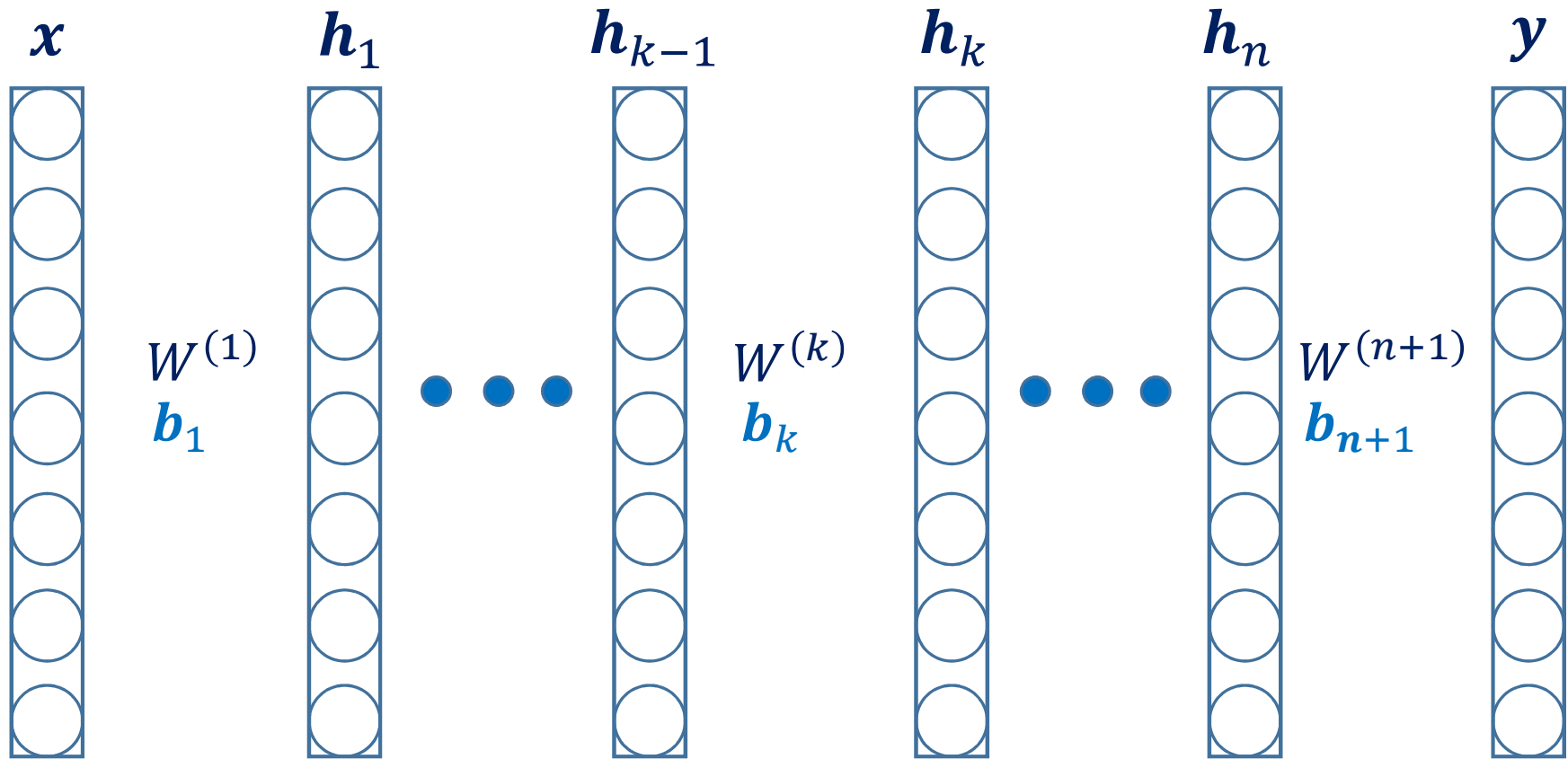
MLP: Computations

$$D = 1000, Q = 500, M = 20$$

OPS	MUL	ADD	DIV	EXP
$\tilde{U} = W^{(1)}X$	500000	500000	0	0
$U = \tilde{U} + \mathbf{b}_1$	0	500	0	0
$H = \text{Sigmoid}(U)$	0	500	500	500
$\tilde{V} = W^{(2)}H$	10000	10000	0	0
$V = \tilde{V} + \mathbf{b}_2$	0	20	0	0
$Y = \text{SoftMax}(V)$	0	20	20	20
TOTAL	510000	511020	520	520

~0.51 MFLOPS

Deep Networks

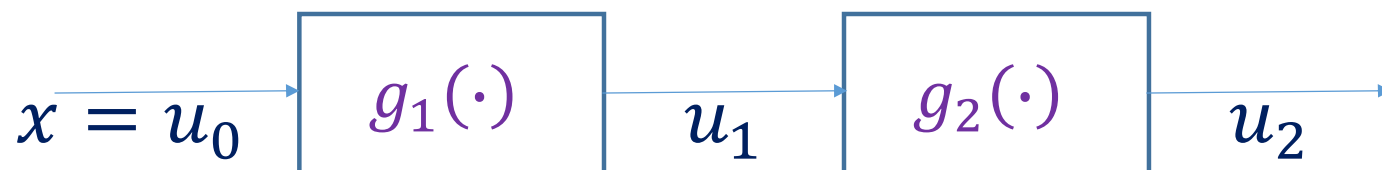


Layers Do Not Necessarily Have Same Number of Nodes

$n > 2$

MLP: Linear Transforms in Layers

$$u_i = g_i(u_{i-1}) = a_i u_{i-1} + b_i$$



$$u_1 = g_1(u_0) = a_1 x + b_1$$

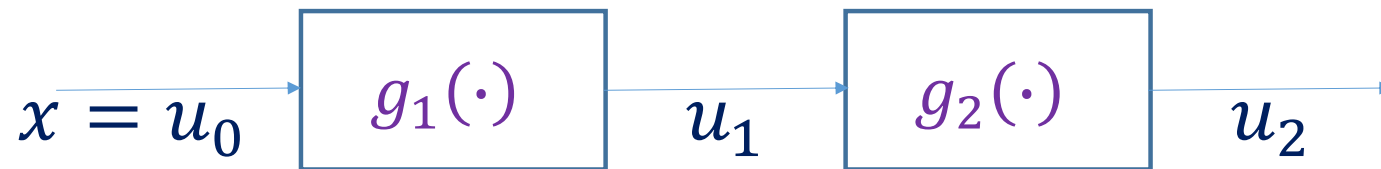
$$u_2 = g_2(u_1) = a_2 u_1 + b_2$$

$$u_2 = a_2(a_1 x + b_1) + b_2 = (a_2 a_1)x + (a_2 b_1 + b_2)$$

Successive Linear or First Order Blocks Lead to an Effective Linear or First Order Block

MLP: Non-Linear Transformations in Layers

$$u_i = g_i(u_{i-1}) = a_i u_{i-1}^2 + b_i u_{i-1} + c_i$$



$$u_1 = g_1(u_0) = a_1 x^2 + b_1 x + c_1$$

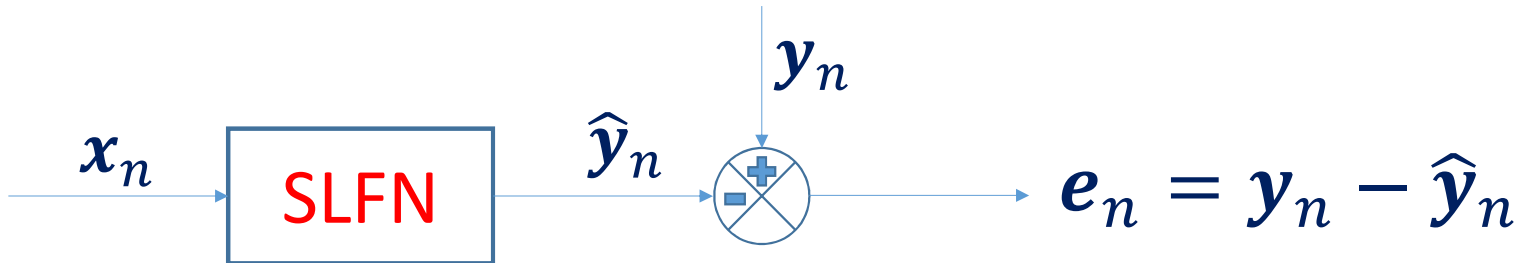
$$u_2 = g_2(u_1) = a_2 u_1^2 + b_2 u_1 + c_2$$

$$u_2 = a_2 (a_1 x^2 + b_1 x + c_1)^2 + b_2 (a_1 x^2 + b_1 x + c_1) + c_2$$

$$u_2 = Ax^4 + Bx^3 + Cx^2 + Dx + E$$

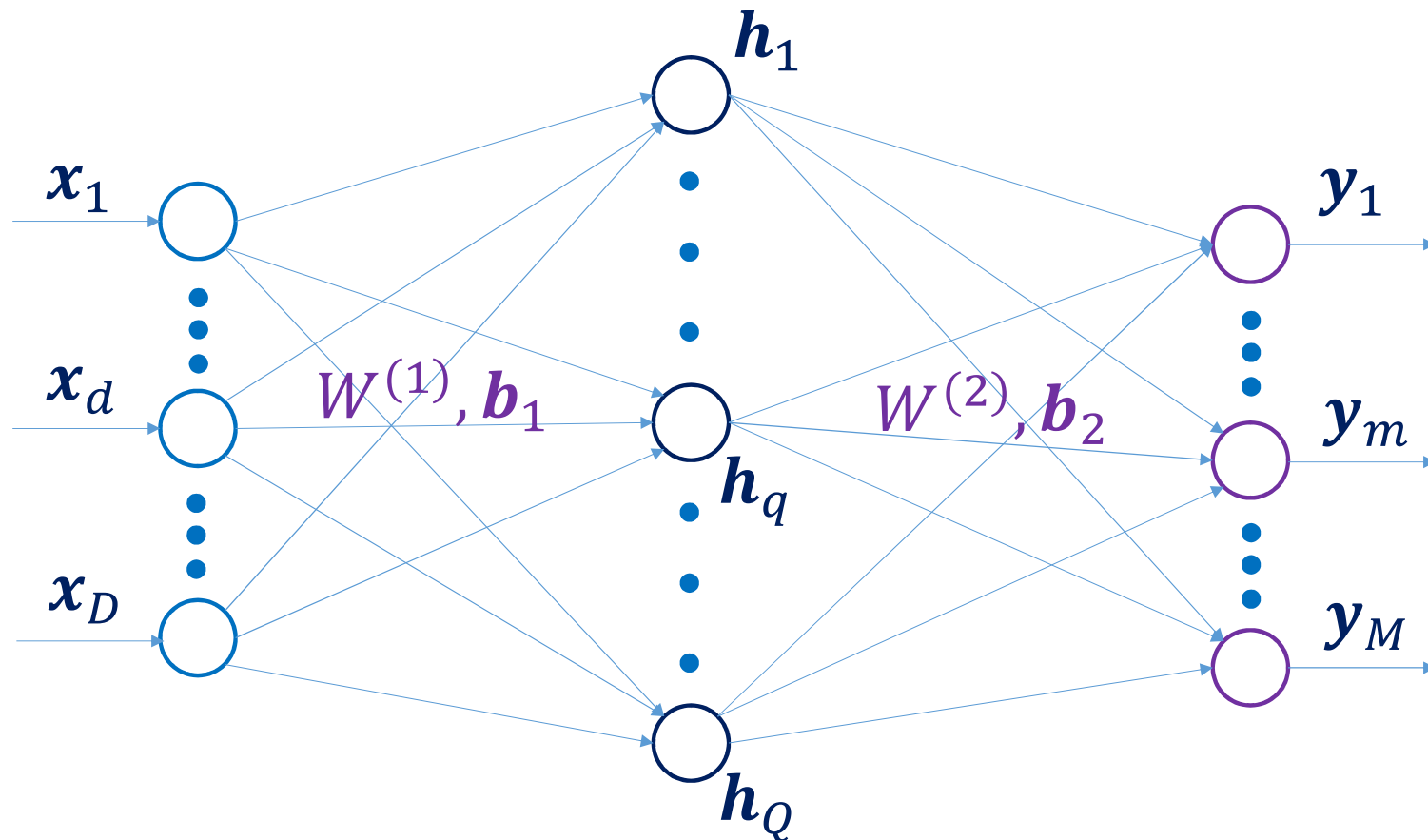
Successive Non-Linear Transforms Lead to Higher Order Non-Linearities

Learning MLP



Minimize Error $\longrightarrow E = \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^M (e_n[m])^2$

Multi-Layered Perceptron (MLP) Architecture



Single Layer Feed Forward Network (SLFN)

The Error Back Propagation Algorithm

Training Dataset

$$\mathcal{S} = \{(\mathbf{x}_n, \mathbf{y}_n); n = 1, \dots, N\}$$

$$\mathbf{x} \in \mathbb{R}^D$$

$$\mathbf{y} \in \mathbb{R}^M$$

$$E = \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^M e_n^2[m]$$

$$W_{ij}^{(2)}(t+1) = W_{ij}^{(2)}(t) - \eta_{ij}^{(t)} \frac{\partial E}{\partial W_{ij}^{(2)}}$$

$$W_{kl}^{(1)}(t+1) = W_{kl}^{(1)}(t) - \eta_{kl}^{(t)} \frac{\partial E}{\partial W_{kl}^{(1)}}$$

$$b_i^{(2)}(t+1) = b_i^{(2)}(t) - \eta_i^{(t)} \frac{\partial E}{\partial b_i^{(2)}}$$

$$b_k^{(1)}(t+1) = b_k^{(1)}(t) - \eta_k^{(t)} \frac{\partial E}{\partial b_k^{(1)}}$$

The Error Back Propagation Algorithm

$$E = \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^M e_n^2[m] \quad \longrightarrow \quad \frac{\partial E}{\partial W_{ij}^{(2)}} = \frac{1}{N} \sum_{n=1}^N 2e_n[i] \frac{\partial e_n[i]}{\partial W_{ij}^{(2)}}$$

$$e_n[i] = y_n[i] - \hat{y}_n[i] \quad \longrightarrow \quad \frac{\partial e_n[i]}{\partial W_{ij}^{(2)}} = - \frac{\partial \hat{y}_n[i]}{\partial W_{ij}^{(2)}}$$

$$\frac{\partial \hat{y}_n[i]}{\partial W_{ij}^{(2)}} = \frac{\partial \hat{y}_n[i]}{\partial v_n[i]} \times \frac{\partial v_n[i]}{\partial W_{ij}^{(2)}}$$

The Error Back Propagation Algorithm

$$\hat{\mathbf{y}}_n[i] = g_2(v_n[i]) \quad \longrightarrow \quad \frac{\partial \hat{\mathbf{y}}_n[i]}{\partial v_n[i]} = g_2^{(1)}(v_n[i])$$

$$v_n[i] = \sum_{q=1}^Q W_{iq}^{(2)} \mathbf{h}_n[q] + b_2[i] \quad \longrightarrow \quad \frac{\partial v_n[i]}{\partial W_{ij}^{(2)}} = \mathbf{h}_n[j]$$

$$\frac{\partial E}{\partial W_{ij}^{(2)}} = \frac{1}{N} \sum_{n=1}^N 2\mathbf{e}_n[i] \frac{\partial \mathbf{e}_n[i]}{\partial \hat{\mathbf{y}}_n[i]} \times \frac{\partial \hat{\mathbf{y}}_n[i]}{\partial v_n[i]} \times \frac{\partial v_n[i]}{\partial W_{ij}^{(2)}}$$

The Error Back Propagation Algorithm

$$\frac{\partial E}{\partial W_{ij}^{(2)}} = -\frac{2}{N} \sum_{n=1}^N \mathbf{e}_n[i] g_2^{(1)}(\mathbf{v}_n[i]) \mathbf{h}_n[j]$$

$$\frac{\partial E}{\partial b_i^{(2)}} = -\frac{2}{N} \sum_{n=1}^N \mathbf{e}_n[i] g_2^{(1)}(\mathbf{v}_n[i])$$

The Error Back Propagation Algorithm

$$E = \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^M e_n^2[m] \quad \longrightarrow \quad \frac{\partial E}{\partial W_{kl}^{(1)}} = \frac{1}{N} \sum_{n=1}^N \sum_{m=1}^M 2e_n[m] \frac{\partial e_n[m]}{\partial W_{kl}^{(1)}}$$

$$e_n[m] = \mathbf{y}_n[m] - \hat{\mathbf{y}}_n[m] \quad \longrightarrow \quad \frac{\partial e_n[m]}{\partial W_{kl}^{(1)}} = - \frac{\partial \hat{\mathbf{y}}_n[m]}{\partial W_{kl}^{(1)}}$$

$$\frac{\partial \hat{\mathbf{y}}_n[m]}{\partial W_{kl}^{(1)}} = \frac{\partial \hat{\mathbf{y}}_n[m]}{\partial \mathbf{v}_n[m]} \times \frac{\partial \mathbf{v}_n[m]}{\partial W_{kl}^{(1)}}$$

The Error Back Propagation Algorithm

$$\hat{\mathbf{y}}_n[m] = g_2(v_n[m]) \longrightarrow \frac{\partial \hat{\mathbf{y}}_n[m]}{\partial v_n[m]} = g_2^{(1)}(v_n[m])$$

$$v_n[m] = \sum_{q=1}^Q W_{mq}^{(2)} \mathbf{h}_n[q] + b_2[m] \longrightarrow \frac{\partial v_n[m]}{\partial W_{kl}^{(1)}} = W_{mk}^{(2)} \frac{\partial h_n[k]}{\partial W_{kl}^{(1)}}$$

$$\frac{\partial E}{\partial W_{kl}^{(1)}} = -\frac{2}{N} \sum_{n=1}^N \sum_{m=1}^M e_n[m] g_2^{(1)}(v_n[m]) W_{mk}^{(2)} \frac{\partial h_n[k]}{\partial W_{kl}^{(1)}}$$

The Error Back Propagation Algorithm

$$\mathbf{h}_n[k] = g_1(\mathbf{u}_n[k]) \quad \longrightarrow \quad \frac{\partial \mathbf{h}_n[k]}{\partial \mathbf{u}_n[k]} = g_1^{(1)}(\mathbf{u}_n[k])$$

$$\mathbf{u}_n[k] = \sum_{d=1}^D W_{kd}^{(1)} \mathbf{x}_n[d] + b_1[k] \quad \longrightarrow \quad \frac{\partial \mathbf{u}_n[k]}{\partial W_{kl}^{(1)}} = \mathbf{x}_n[l]$$

$$\frac{\partial \mathbf{h}_n[k]}{\partial W_{kl}^{(1)}} = \frac{\partial \mathbf{h}_n[k]}{\partial \mathbf{u}_n[k]} \times \frac{\partial \mathbf{u}_n[k]}{\partial W_{kl}^{(1)}} = g_1^{(1)}(\mathbf{u}_n[k]) \mathbf{x}_n[l]$$

The Error Back Propagation Algorithm

$$\frac{\partial E}{\partial W_{kl}^{(1)}} = -\frac{2}{N} \sum_{n=1}^N \sum_{m=1}^M e_n[m] g_2^{(1)}(\mathbf{v}_n[m]) W_{mk}^{(2)} g_1^{(1)}(\mathbf{u}_n[k]) \mathbf{x}_n[l]$$

$$\frac{\partial E}{\partial b_k^{(1)}} = -\frac{2}{N} \sum_{n=1}^N \sum_{m=1}^M e_n[m] g_2^{(1)}(\mathbf{v}_n[m]) W_{mk}^{(2)} g_1^{(1)}(\mathbf{u}_n[k])$$

Loss Functions

Zero-One Loss

$$E = \frac{1}{N} \sum_{n=1}^N \{1 - \delta[y_n - \hat{y}_n]\}$$

Quadratic Loss

$$E = \frac{1}{N} \sum_{n=1}^N \|\mathbf{y}_n - \hat{\mathbf{y}}_n\|_2^2$$

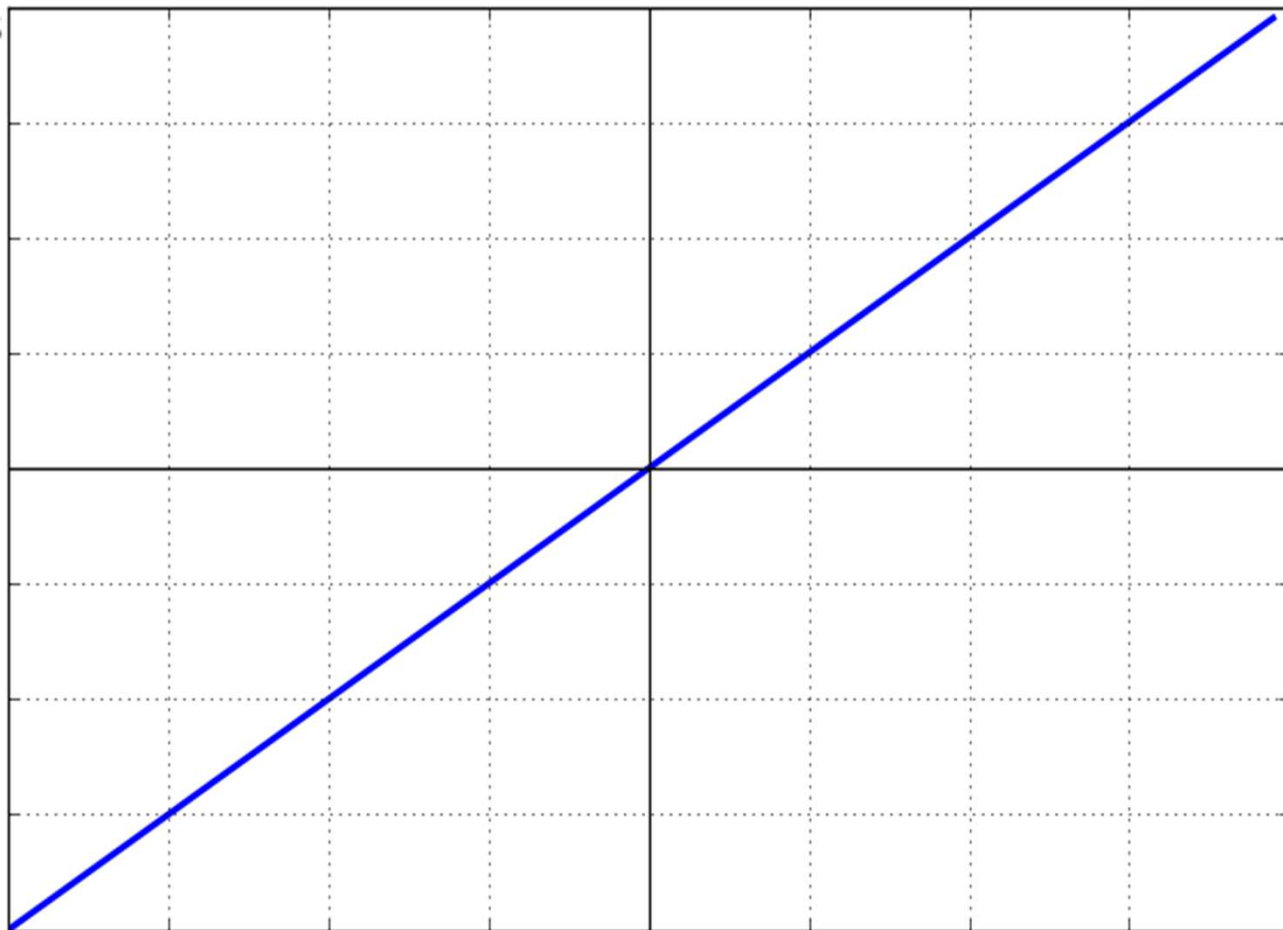
L1 Loss

$$E = \frac{1}{N} \sum_{n=1}^N \|\mathbf{y}_n - \hat{\mathbf{y}}_n\|_1$$

Cross-Entropy Loss

$$E = - \sum_{n=1}^N \sum_{m=1}^M \mathbf{y}_n[m] \log \hat{\mathbf{y}}_n[m]$$

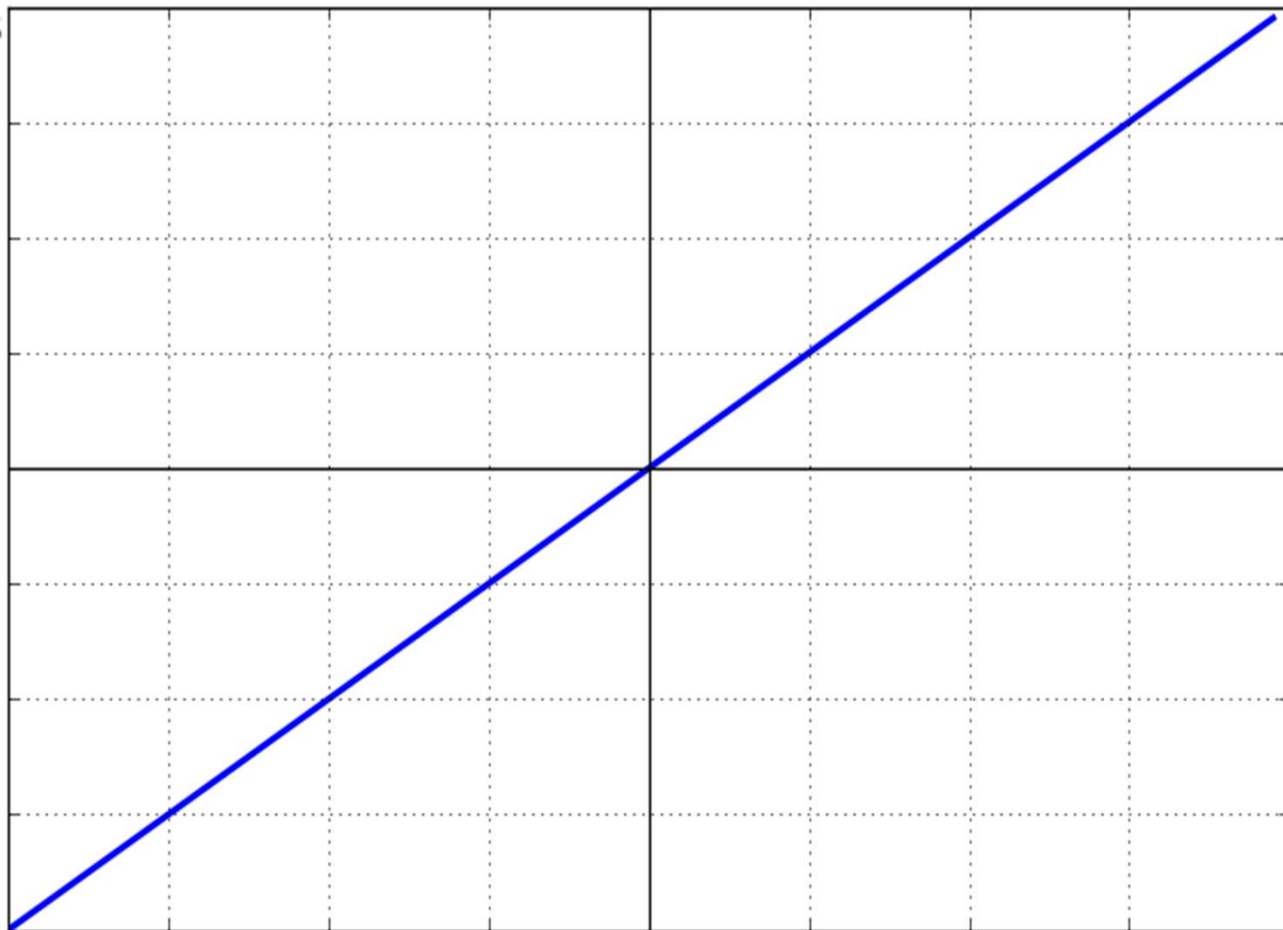
Activation Functions: Linear



$$g(u) = u$$

$$g'(u) = ?$$

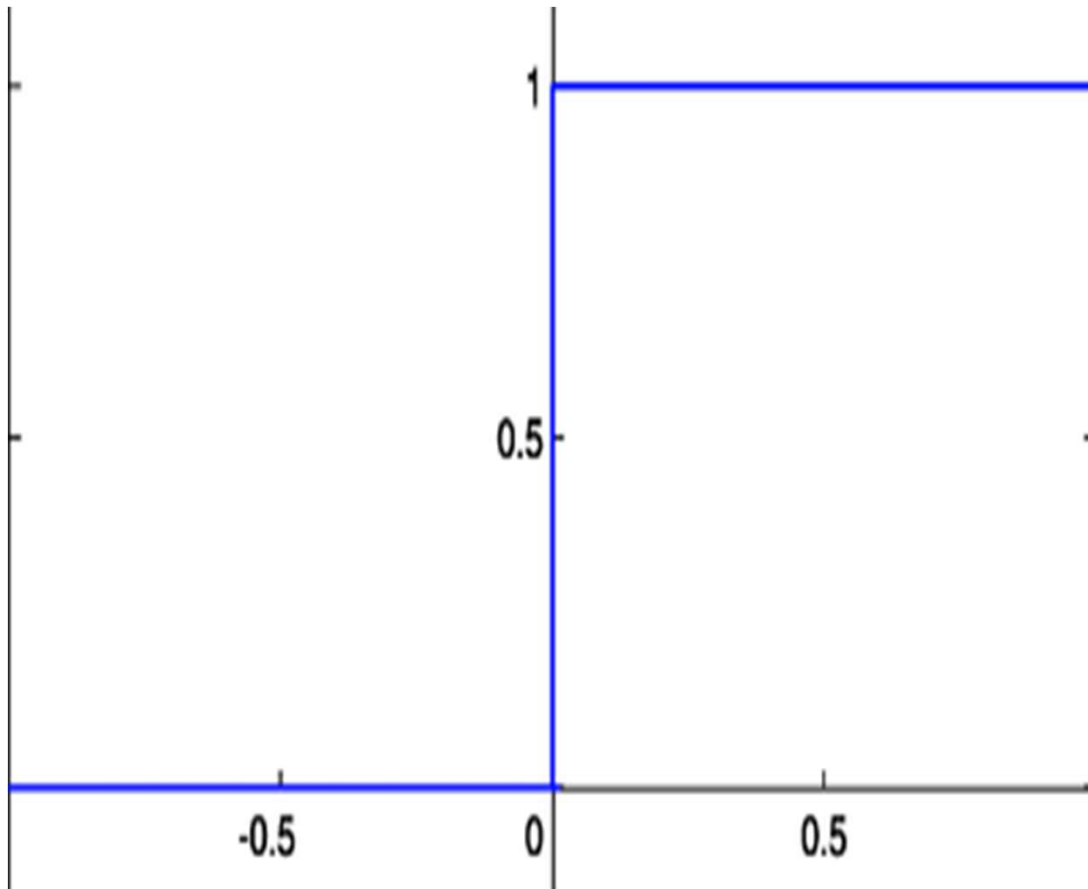
Activation Functions: Linear



$$g(u) = u$$

$$g'(u) = 1$$

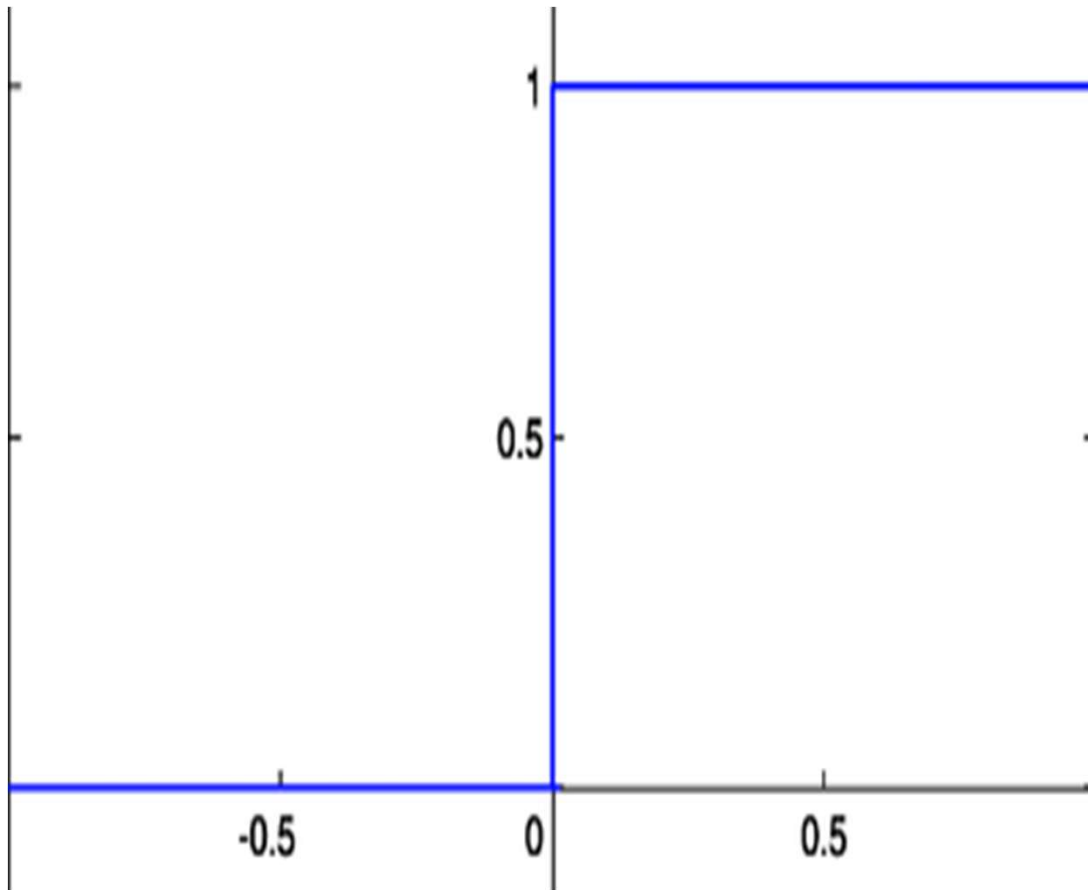
Activation Functions: Unit Step



$$g(u) = \begin{cases} 1, & u \geq 0 \\ 0, & u < 0 \end{cases}$$

$$g'(u) = ?$$

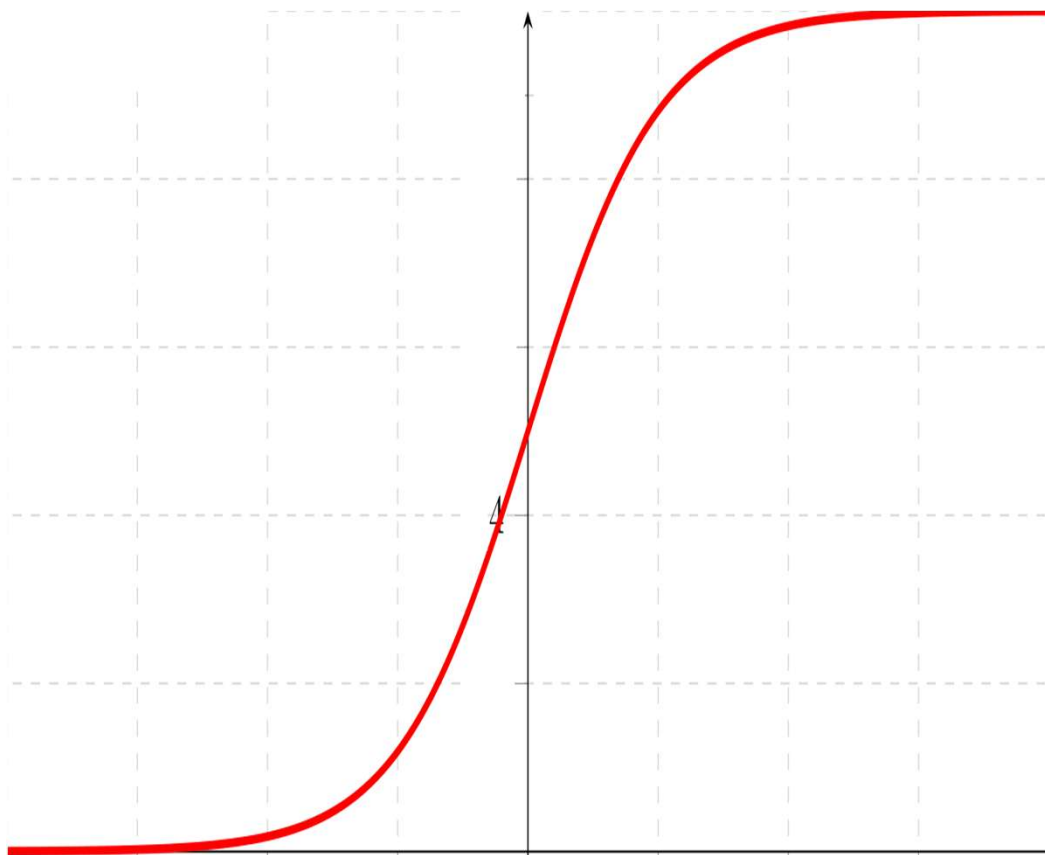
Activation Functions: Unit Step



$$g(u) = \begin{cases} 1, & u \geq 0 \\ 0, & u < 0 \end{cases}$$

$$g'(u) = \begin{cases} 0, & u \neq 0 \\ \neg D, & u = 0 \end{cases}$$

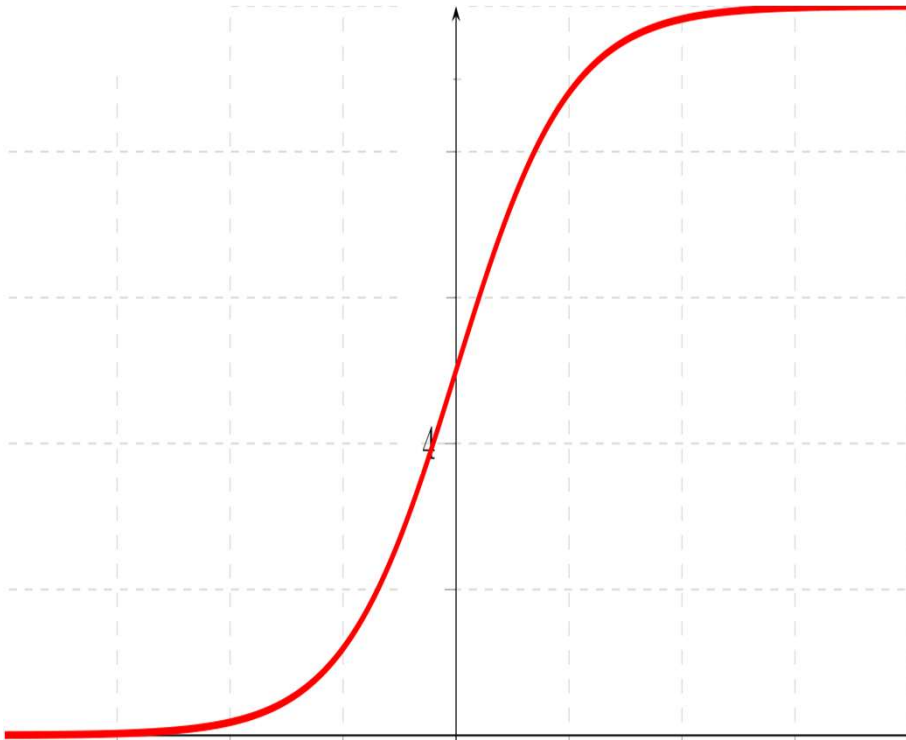
Activation Functions: Sigmoid



$$g(u) = \frac{1}{1 + e^{-u}}$$

$$g'(u) = ?$$

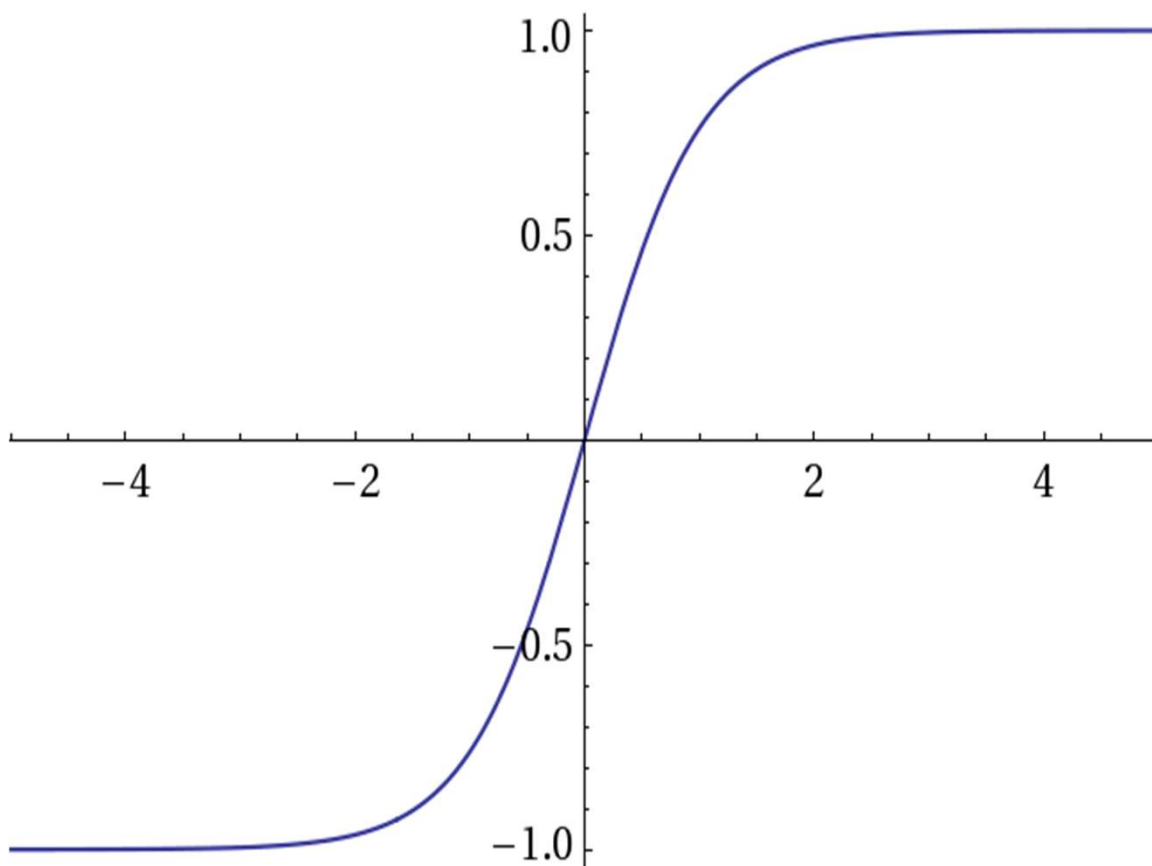
Activation Functions: Sigmoid



$$g(u) = \frac{1}{1 + e^{-u}}$$

$$g'(u) = g(u)\{1 - g(u)\}$$

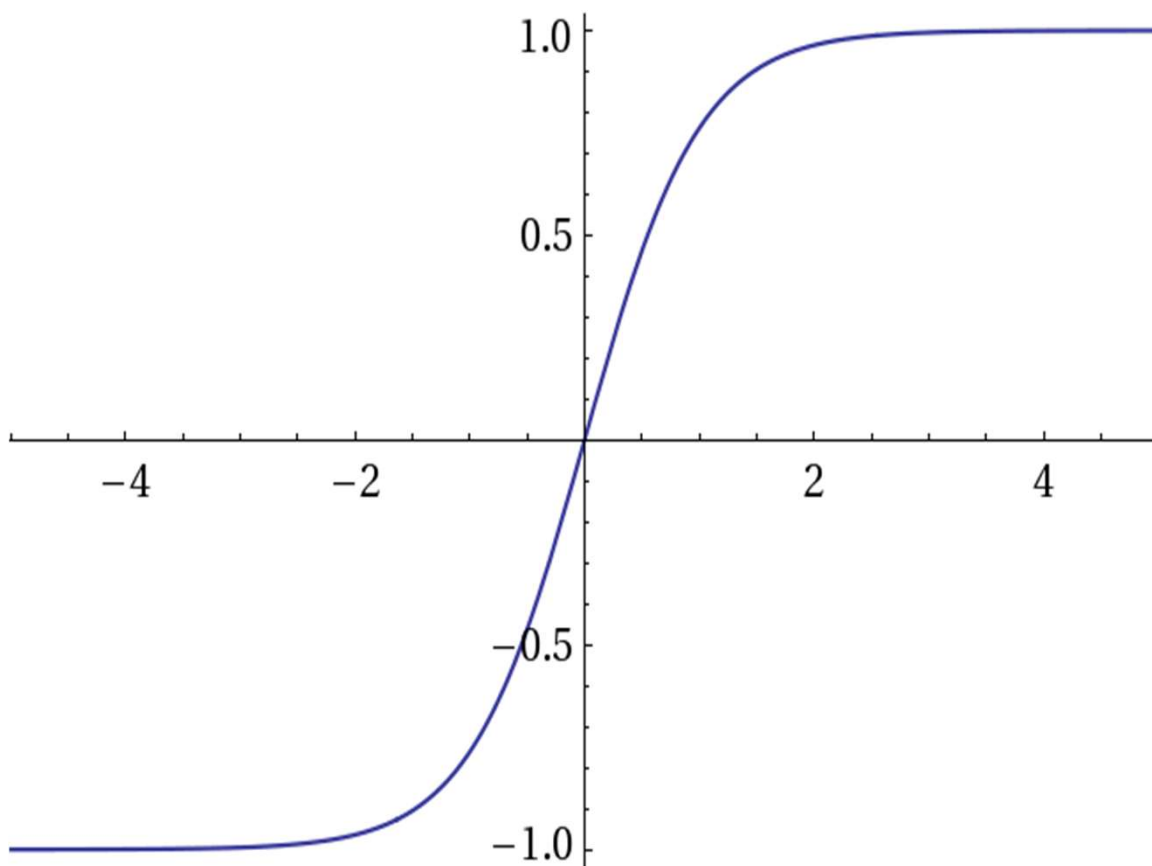
Activation Functions: TanH



$$g(u) = \frac{e^u - e^{-u}}{e^u + e^{-u}}$$

$$g'(u) = ?$$

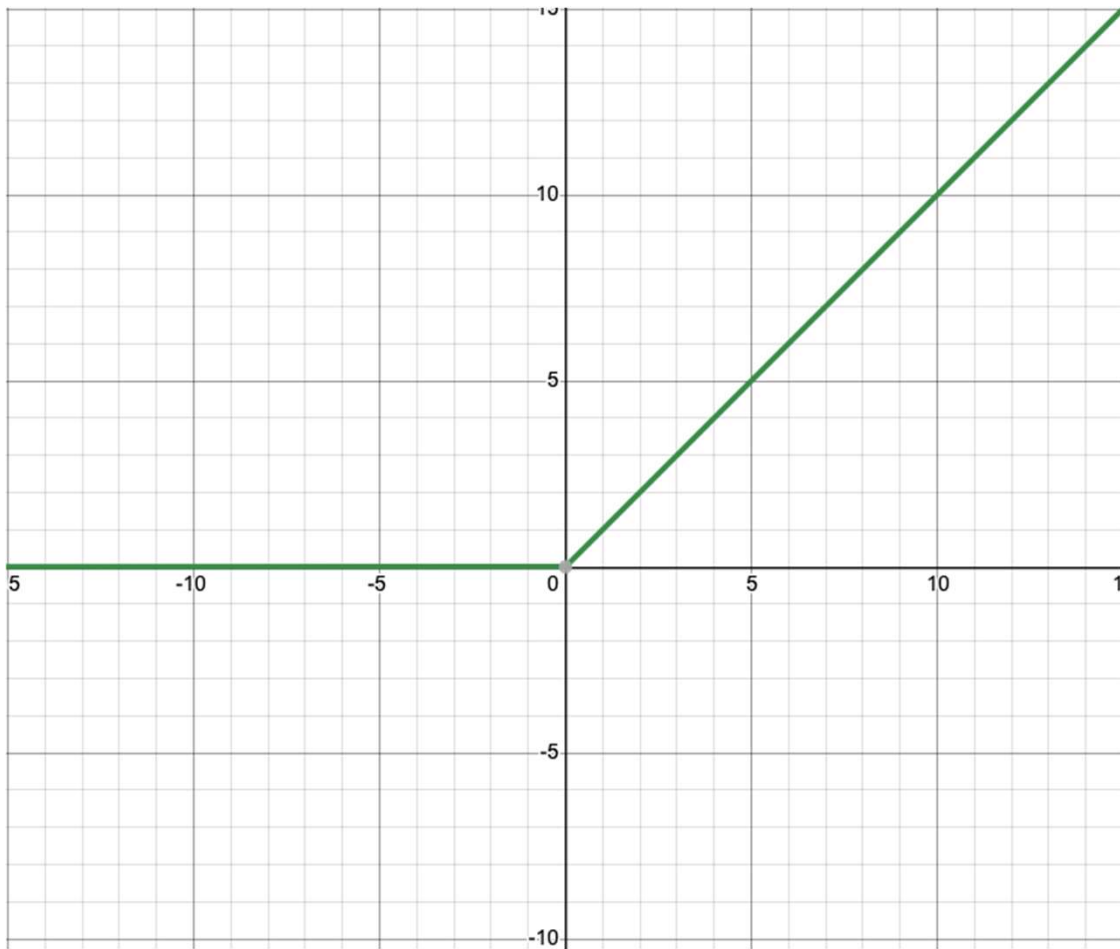
Activation Functions: TanH



$$g(u) = \frac{e^u - e^{-u}}{e^u + e^{-u}}$$

$$g'(u) = 1 - \{g(u)\}^2$$

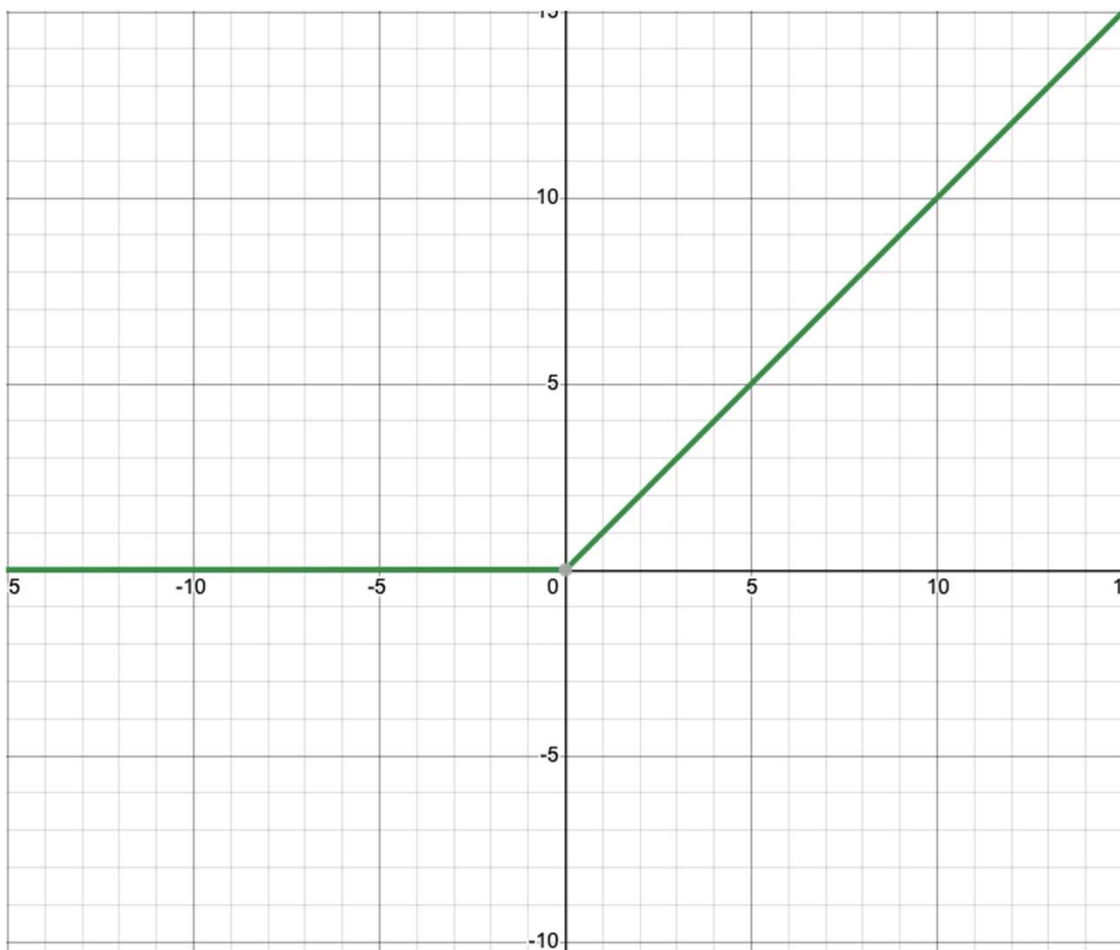
Activation Functions: ReLU



$$g(u) = \begin{cases} u, & u \geq 0 \\ 0, & u < 0 \end{cases}$$

$$g'(u) = ?$$

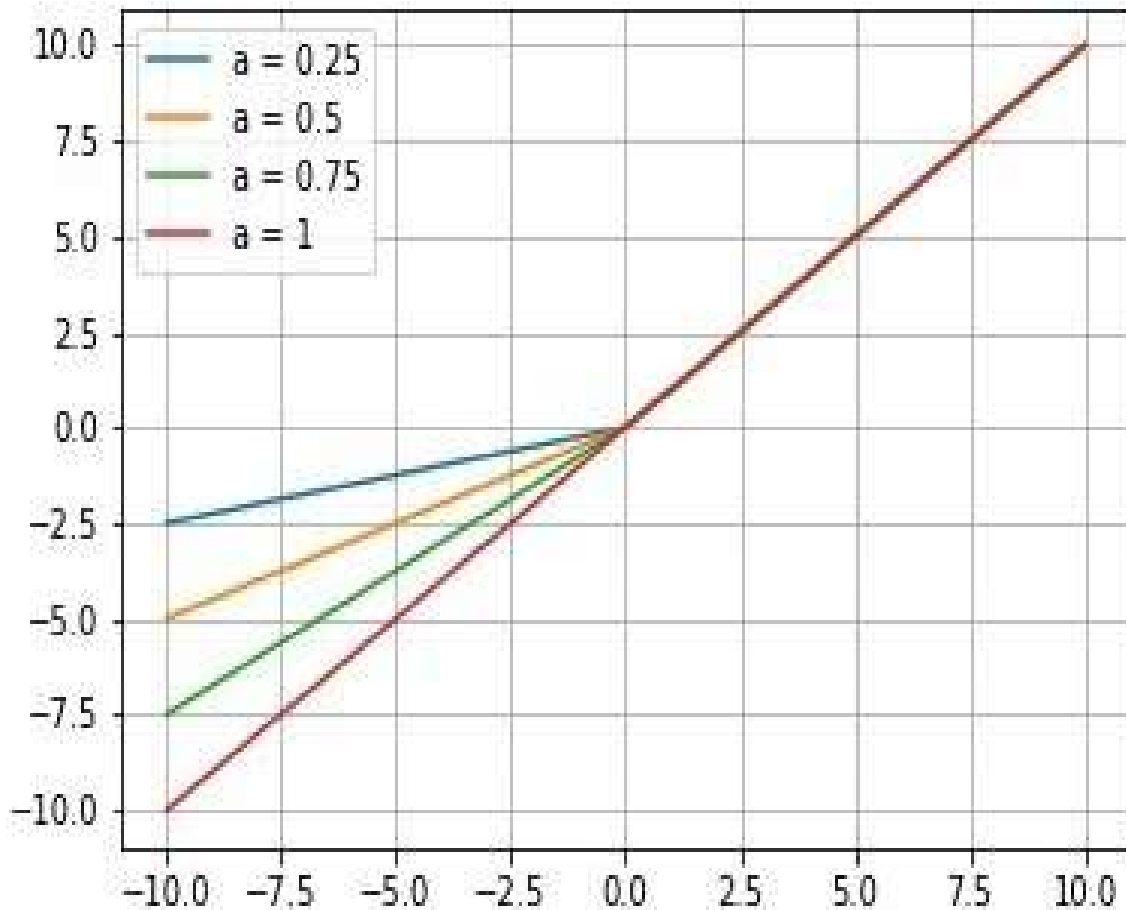
Activation Functions: ReLU



$$g(u) = \begin{cases} u, & u \geq 0 \\ 0, & u < 0 \end{cases}$$

$$g'(u) = \begin{cases} 1, & u \geq 0 \\ 0, & u < 0 \end{cases}$$

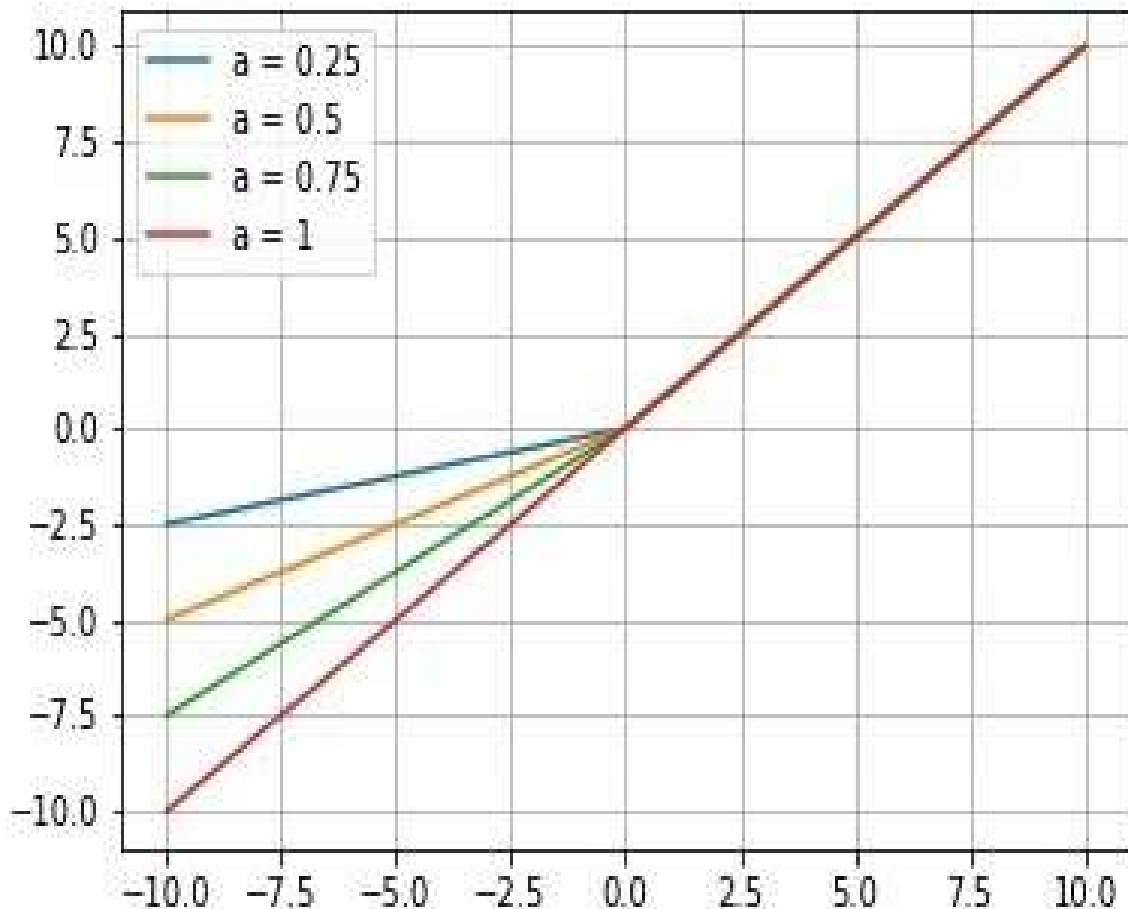
Activation Functions: PReLU



$$g(u) = \begin{cases} u, & u \geq 0 \\ au, & u < 0 \end{cases}$$

$$g'(u) = ?$$

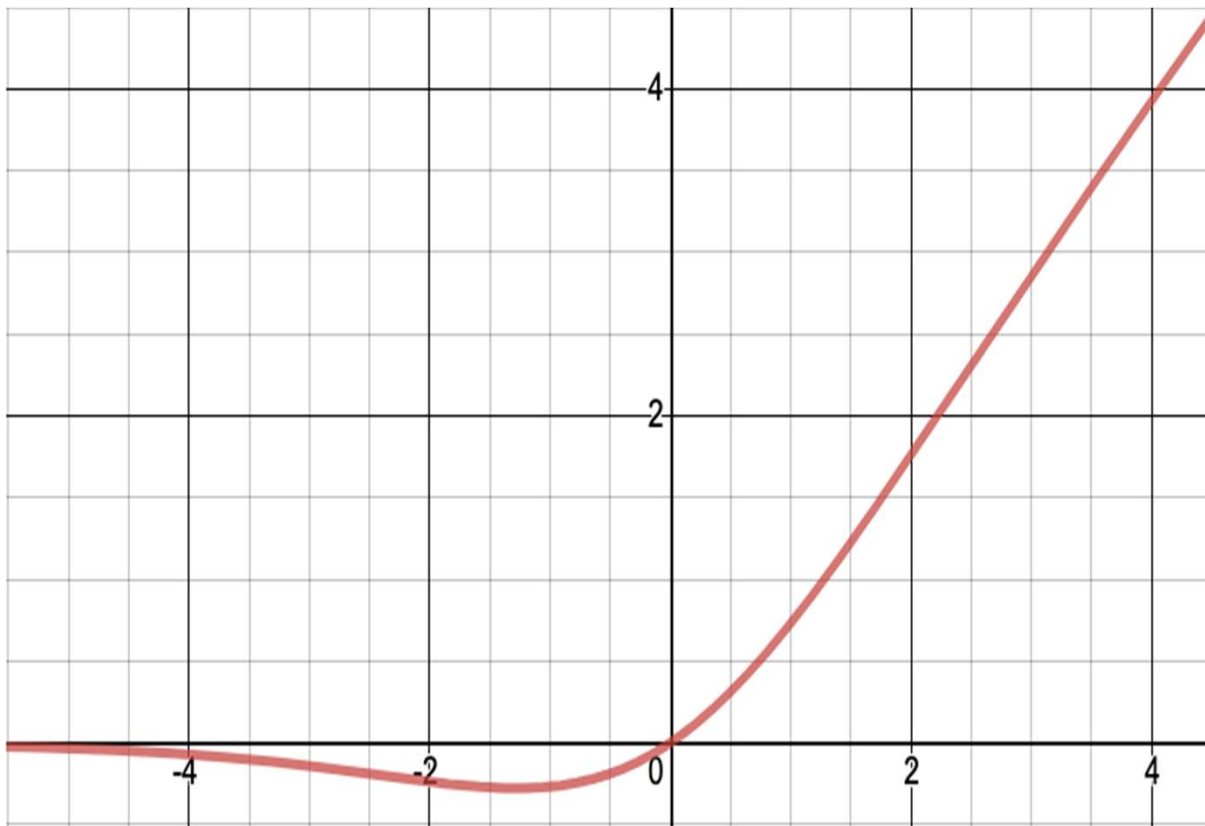
Activation Functions: PReLU



$$g(u) = \begin{cases} u, & u \geq 0 \\ au, & u < 0 \end{cases}$$

$$g'(u) = \begin{cases} 1, & u \geq 0 \\ a, & u < 0 \end{cases}$$

Activation Functions: Swish

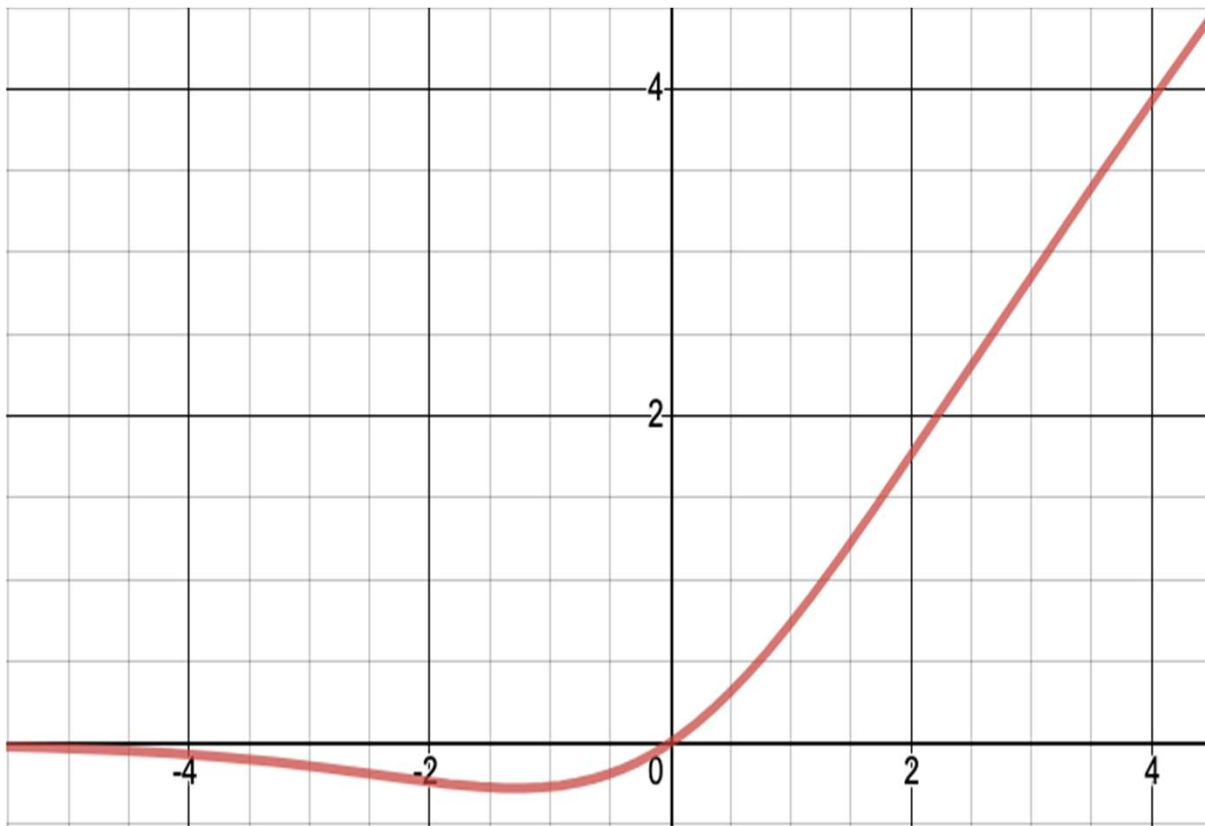


$$g_s(u) = \frac{1}{1 + e^{-u}}$$

$$g(u) = u g_s(u)$$

$$g'(u) = ?$$

Activation Functions: Swish



$$g_s(u) = \frac{1}{1 + e^{-u}}$$

$$g(u) = u g_s(u)$$

$$g'(u) = g_s(u) + g(u)\{1 - g_s(u)\}$$

Feed Forward through SLFN

$$\mathcal{S} = \{(\mathbf{x}_n, \mathbf{y}_n); n = 1, \dots, N\}$$


$$\mathbf{x} \in \mathbb{R}^D$$

$$\mathbf{y} \in \mathbb{R}^M$$

$$X = [\mathbf{x}_1 \quad \cdots \quad \mathbf{x}_n \quad \cdots \quad \mathbf{x}_N]_{D \times N} \quad U = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_n \quad \cdots \quad \mathbf{u}_N]_{Q \times N}$$

$$H = [\mathbf{h}_1 \quad \cdots \quad \mathbf{h}_n \quad \cdots \quad \mathbf{h}_N]_{Q \times N} \quad V = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n \quad \cdots \quad \mathbf{v}_N]_{M \times N}$$

$$Y = [\mathbf{y}_1 \quad \cdots \quad \mathbf{y}_n \quad \cdots \quad \mathbf{y}_N]_{M \times N}$$

Extreme Learning Machines (ELM)

The Parameters $\{W^{(1)}, \mathbf{b}_1\}$ are Not Learned from Data but are Randomly Generated in the interval $\left[-\frac{2.4}{Q}, \frac{2.4}{Q}\right]$ (experimentally established result)

$$U = [W^{(1)} \quad \mathbf{b}_1] \begin{bmatrix} X \\ \mathbf{1} \end{bmatrix}$$



$$H = g_1(U)$$

Any Random Parameter Set $\{W^{(1)}, \mathbf{b}_1\}$ will generate Different H from X

Extreme Learning Machines (ELM)

Assumption: Inverse of the activation function $g_2(\cdot)$ of output layer exists

Evaluate: $\mathbf{v}_n[m] = g_2^{-1}(\mathbf{y}_n[m]); m = 1 \dots M; n = 1, \dots N$

Construct: $V = [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n \quad \cdots \quad \mathbf{v}_N]_{M \times N}$

$$P = [W^{(2)} \quad \mathbf{b}_2]_{M \times (Q+1)}$$

Hidden-To-Output Layer Parameters

Hidden Layer Activations

$$A = \begin{bmatrix} \mathbf{h}_1 & \cdots & \mathbf{h}_n & \cdots & \mathbf{h}_N \\ 1 & \cdots & 1 & \cdots & 1 \end{bmatrix}_{(Q+1) \times N}$$

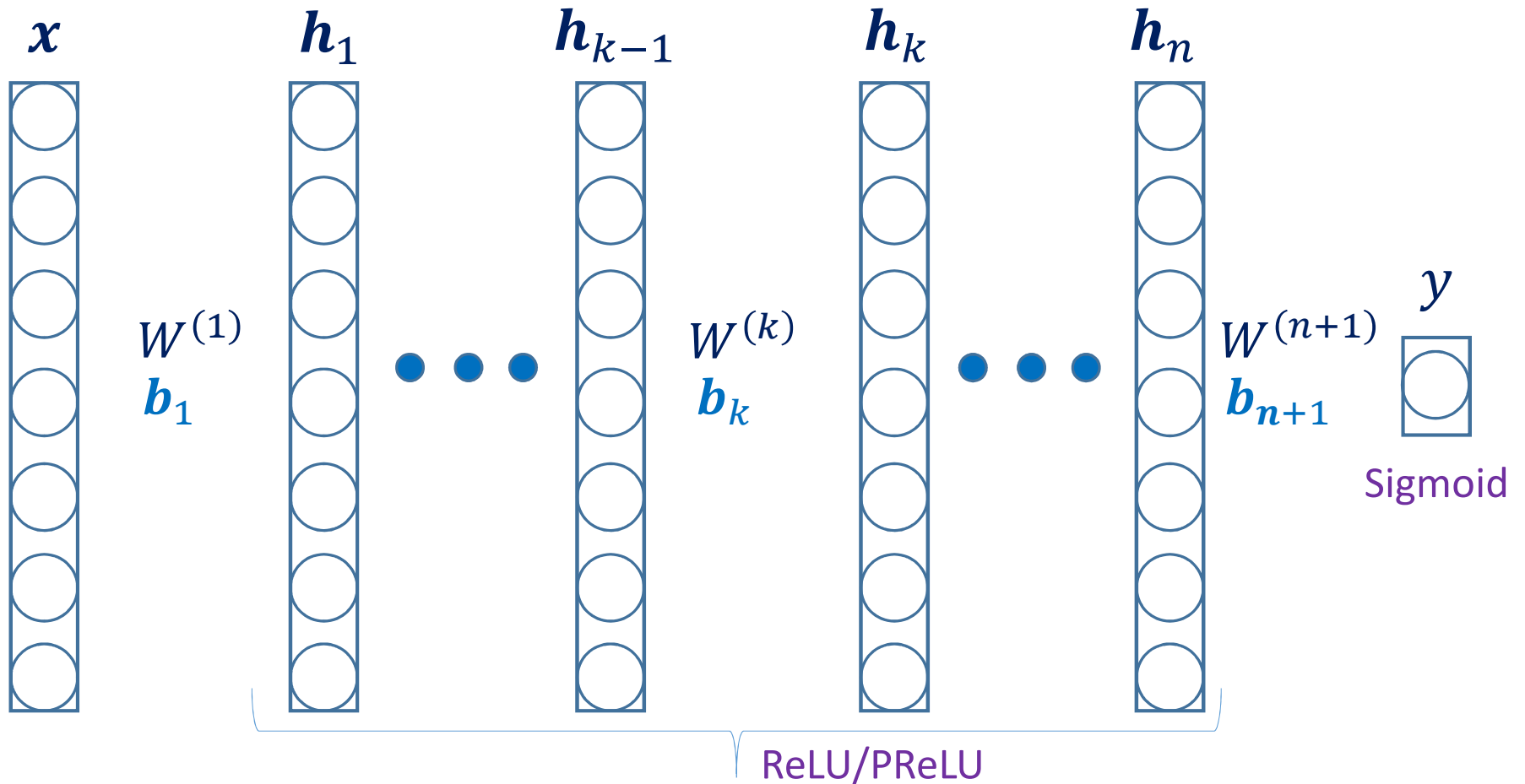
$$P = V A^T (A A^T)^{-1}$$

Moore-Penrose Generalized Pseudo-Inverse

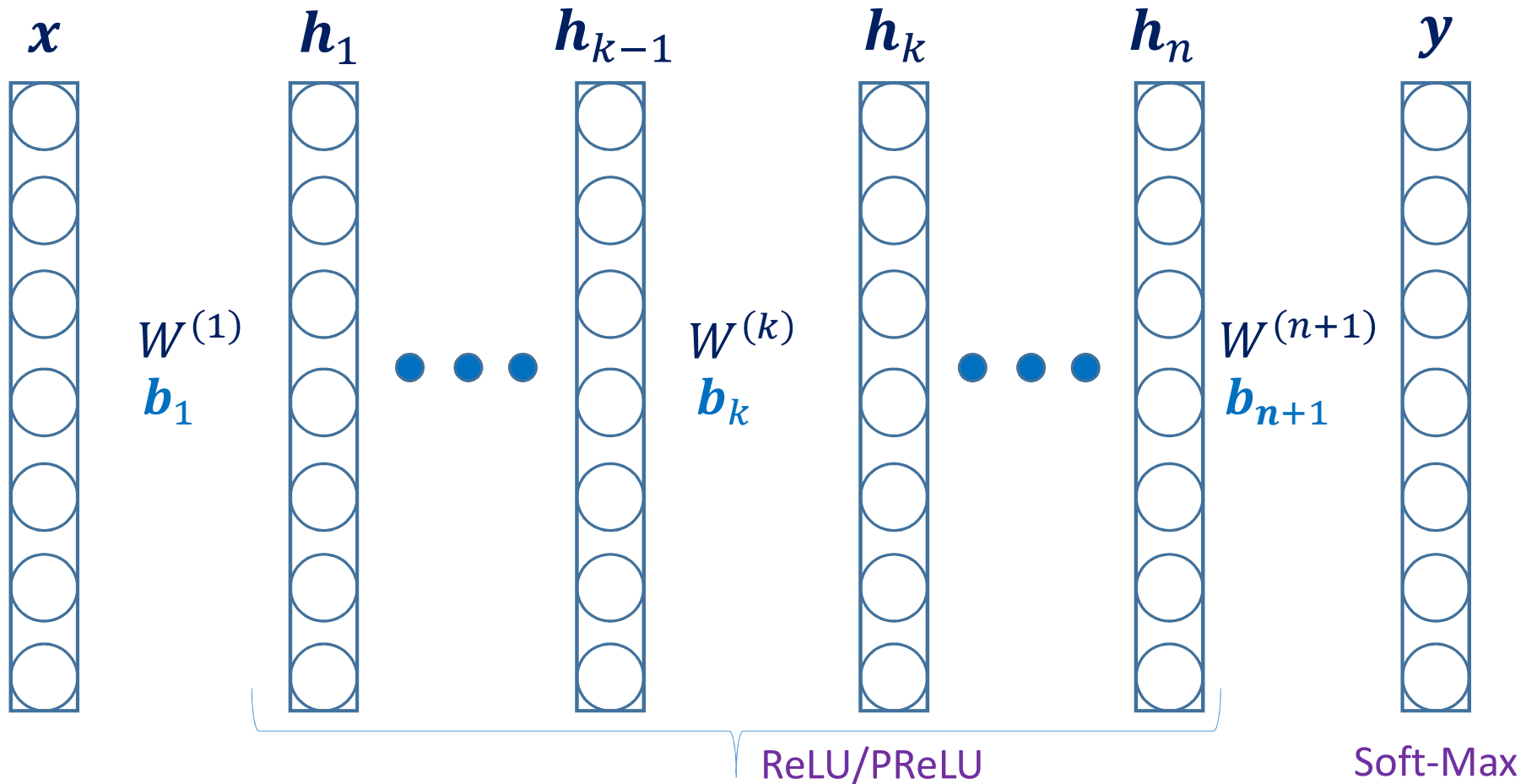
MLP Applications

- Binary and Multi-Category Classification Problems
- Regression Problems
- Learning Similarity Measures
- Non-linear Dimensionality Reduction

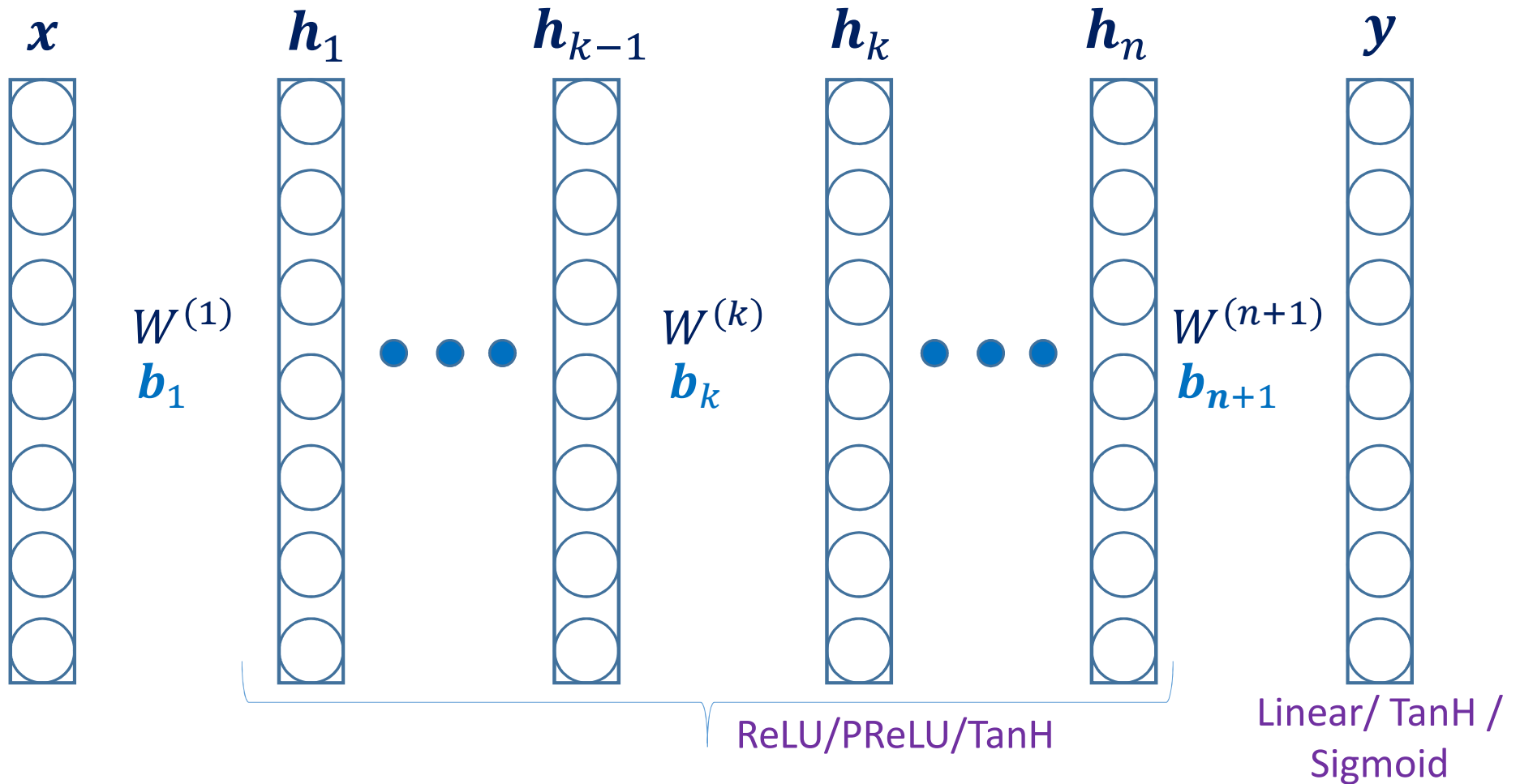
MLP: Binary Classification Problem



MLP: Multi-Category Classification

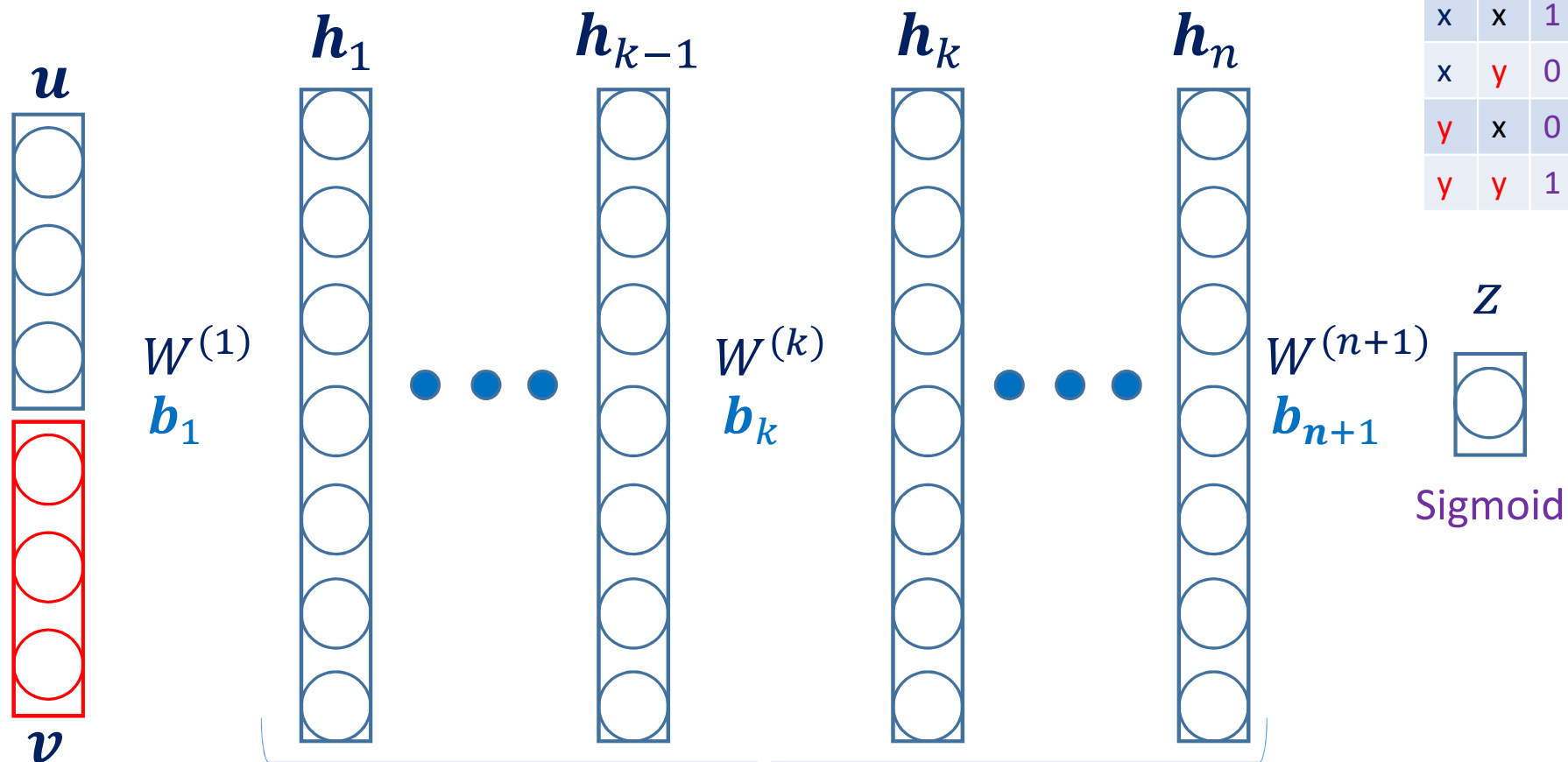


MLP: Regression Problems



MLP: Learning Similarity Measures

u	v	z
x	x	1
x	y	0
y	x	0
y	y	1

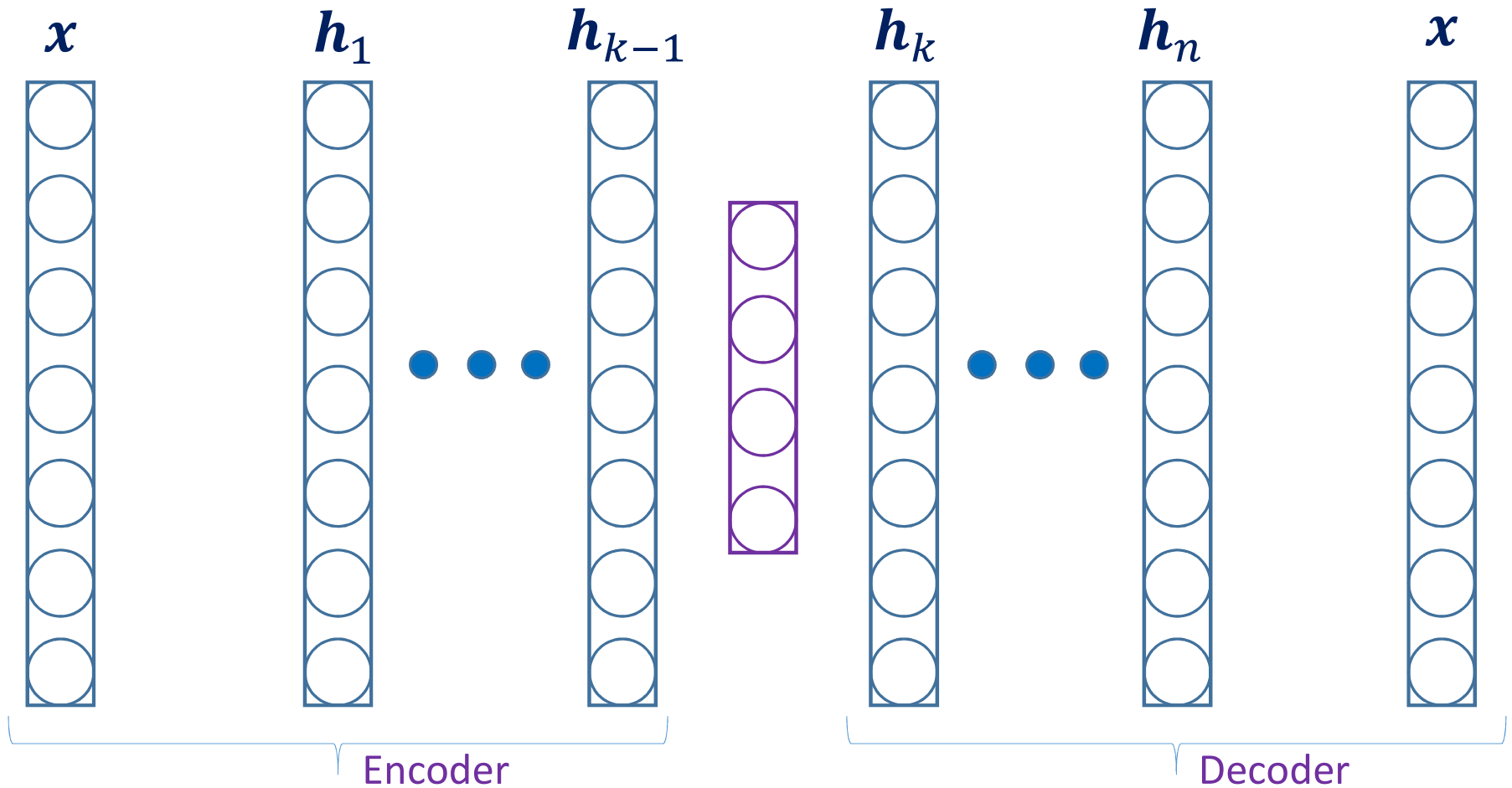


$$\mathcal{S}_A = \{x_n : x_n \in \mathbb{R}^d; n = 1, \dots, N\}$$

ReLU/PReLU

$$\mathcal{S}_B = \{y_m : y_m \in \mathbb{R}^d; m = 1, \dots, M\}$$

MLP: Non-Linear Dimensionality Reduction



Summary

- Separability of Data in Feature Space
- Multi-Layered Perceptron (MLP) Architecture
- Feed Forward through MLP
- Interpretation of Layers in Multi-layered Networks
- Error Backpropagation Algorithm
- Activation Functions and Loss Functions
- Extreme Learning Machines
- MLP Applications



Thank You